

BRAC IT CodeSprint 2026

Project BRD

MathKotha - মাঠকথা

Turning field officers' voices into an early-warning system for microfinance

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3. Problem Statement

BRAC Microfinance serves 8 million+ borrowers through 2,900+ branches and disbursed USD 6.6 billion in 2025, 89% of it to women. The system records whether an instalment was collected. It has no way to record why it was not — and the one person who knows is never asked.

3.1 What is being lost

- A field officer covers roughly 40 borrower visits a week. At every visit she sees the cause before the ledger does — a flooded market, sick livestock, a shop that has closed, illness in the household, a husband who has lost work.
- The MIS has a field for collection status. It has no field for cause. She records “not collected” and moves on.

3.2 What that costs

- Lost time.** The branch manager learns of the problem 2–4 weeks later from the arrears report — by which point it is a recovery problem, not a prevention one.
- Invisible correlation.** Because causes are never recorded, nine borrowers hit by the same flooded market appear as nine unrelated late payments. A shock that hits an entire ward is never seen as one event.
- No institutional memory.** The reasons a portfolio deteriorates exist only in officers' heads, and leave when they do.

3.3 Why the obvious fix has not worked

- Structured forms have been tried and are quietly skipped. After 40 visits a week, typing a report on a mid-range Android phone is not realistic, and officers are right to skip it.
- Any capture method that costs more than roughly 20 seconds per visit will not be used, whatever the policy says.** That constraint — not the technology — is what the design has to satisfy.

4. Proposed Solution

Replace the form with twenty seconds of speech, then turn what officers already say into a branch-level early-warning signal.

1. SPEAK — 20s Bangla	2. TRANSCRIBE — Bangla ASR	3. EXTRACT — fixed schema	4. CONFIRM — officer, 10s	5. CLUSTER — ward + cause	6. ACT — manager digest
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4.1 Capture — built around the 20-second constraint

- After each visit the officer taps one button and speaks in Bangla, in her own words, the way she would tell a colleague. No form, no dropdown, no typing.
- Offline-first. Audio is queued on the device and syncs when the network returns, so field coverage is never a dependency.

4.2 Processing — speech becomes a record

- Bangla speech-to-text produces a transcript with per-segment confidence.
- A language model extracts a fixed-schema visit record: collection status, cause code from a closed list, cause detail in free text, shock type, borrower sentiment, follow-up date and urgency. Every field carries a confidence, and the model abstains rather than guesses.
- The officer confirms or corrects the record in under ten seconds — nothing is stored unconfirmed — and the confirmed record is written beside the loan record, so the “why” finally sits next to the “what”.

4.3 Intelligence — the layer that does not exist today

- A clustering engine groups confirmed records by branch, ward, cause and time window, and semantically groups free-text causes that share no cause code.
- For each cluster it computes exposure at risk — the summed outstanding balance of the affected borrowers.
- Clusters above threshold become alerts on the branch manager’s morning digest, ranked by money at risk, each drilling down to the borrower list and the original voice notes.

4.5 Boundaries, set deliberately

- The system advises; a human decides. No automated credit decision, no automated rejection, no borrower score that affects lending. Every record is officer-confirmed and attributable, with a full audit trail of who spoke, who confirmed and who acted.

5. Target Users

User	Role in the system	What they get that they do not have now
Field Officer / Programme Organiser	Primary author — speaks the note, confirms the record	Twenty seconds replaces a report she does not write at all today. Her knowledge stops being discarded.
Branch Manager	Primary consumer — reads the digest, decides the action	Sees a correlated shock in her ward inside 24 hours instead of three weeks, with a taka figure attached.
Area, Regional and Head Office teams	Portfolio oversight, risk, product and programme	Cross-branch view of emerging causes, and the first queryable dataset of why repayment fails — by cause, geography and season.
Borrower (indirect)	Beneficiary	Earlier and better-targeted support — rescheduling or advice — instead of pressure after default.

6. Key Features

ID	Feature	Scope
F1	One-tap Bangla voice capture, 30s maximum, offline-first with a sync queue	MVP
F2	Bangla speech-to-text with an editable transcript	MVP
F3	Structured extraction into the fixed visit-record schema (Appendix A)	MVP
F4	Officer confirmation screen — accept or correct in ≤10 seconds	MVP
F5	Cause taxonomy — closed code list plus free-text detail	MVP

F6	Clustering engine — group by branch, ward, cause and time window	MVP
F7	Exposure-at-risk calculation per cluster	MVP
F8	Branch manager digest ranked by money at risk, drilling down to affected borrowers and the original voice notes	MVP
F9	Audit trail on every record — speaker, confirmer, timestamps	MVP
F10	Semantic grouping of free-text causes, and emerging-cause detection that proposes new taxonomy codes	Phase 2
F11	MIS API and export, supervisor quality view, and multi-language extension beyond Bangla	Phase 2

Explicitly out of scope: credit scoring, loan approval or rejection, automated collections, a borrower-facing app, or replacing any existing MIS function.

7. AI Component

Three places, precisely — and one place it is deliberately kept out of.

Where	Model type	What it does, and how it is judged
A. Speech recognition	Bangla ASR — Whisper-class or a Bangla fine-tuned variant	Input: 20–30s of field audio with regional accent and street noise. Output: transcript plus per-segment confidence. Chosen by measured word error rate on the team’s own recorded test set, not by reputation.
B. Structured extraction	Instruction-tuned LLM, few-shot on real transcripts	Input: transcript. Output: JSON against a fixed schema — closed-list classification for cause code, free text for detail, a confidence per field, and explicit abstention when unsure. The closed list is what makes clustering possible downstream.
C. Semantic clustering	Sentence embeddings + nearest-neighbour grouping	Groups free-text causes that mean the same thing in different words — “the market flooded”, “water in the shop”, “bazar shut from water”. Also surfaces recurring causes the taxonomy has no code for yet.

Where AI is deliberately NOT used

No credit scoring. No approval or rejection decision. No borrower ranking that affects lending.

The model produces a draft record — the officer confirms it. The model produces an alert — the manager decides the action. There are two humans in the loop by design, not by omission.

This is a deliberate compliance choice. **Bangladesh’s draft National AI Policy 2026–2030** applies strict liability to high-risk AI systems, and an automated credit decision affecting a low-income borrower would sit squarely in that category. Keeping both decisions with people keeps MathKotha outside it.

8. Expected Outcome / Impact

Change	What it means in practice
Lead time collapses	Cause reaches the branch manager in under 24 hours instead of 2–4 weeks — two to four weeks of earlier action on every case.

Prevention replaces recovery	Rescheduling, advice or a top-up before default costs less than recovery after it, and keeps the client relationship intact.
Correlated shocks become visible	A ward-level event is seen as one event with a taka figure attached. This capability does not exist in the current system in any form.
Capture costs almost nothing	~20 seconds of speech plus ~10 seconds of confirmation, replacing a report nobody writes today. The data is net new, and “why do borrowers here fall behind in this season” becomes a query instead of an opinion.
Value beyond one branch	On a book of this size any sustained reduction in portfolio-at-risk is materially valuable — section 10 measures it rather than asserting it here. The same module is portable to every country the platform runs in, and reusable for any door-to-door operation: community health, programme monitoring, agriculture extension.

9. Data / Resources Required

For the nine-day build — nothing external is needed

- **Voice corpus.** 60–100 Bangla voice notes recorded by the team in the voice of a field officer, varied by dialect, speed and background noise, each with a ground-truth label. Built on days 1–2.
- **Synthetic borrower dataset.** ~500 rows — borrower ID, name, branch, ward, outstanding balance, due date. Generated.
- **Models.** Bangla ASR (Whisper large-v3 or a Bangla fine-tuned variant, open weights); an instruction-tuned LLM via API or run locally; a sentence-embedding model for the Phase 2 clustering feature.
- **Stack.** Mobile client (Flutter, React Native or a PWA), any backend framework, PostgreSQL, object storage for audio.

No BRAC data, no system access and no third-party partner is required to build or demo this. Nothing outside the team’s own control can block the sprint.

For production deployment, after the sprint

- Read access to borrower and loan master data — borrower ID, branch, ward, outstanding balance, due date — plus officer-to-branch mapping.
- A write API into the microfinance MIS for confirmed visit records.
- A cause taxonomy — the closed code list — agreed with the microfinance programme team.
- Audio storage with a defined retention period, a consent and privacy policy covering recordings that mention borrowers, and GPU or hosted inference capacity sized to daily visit volume.

10. Success Criteria

Inside the build round —

ID	Criterion	Target and method
SC1	End-to-end loop works	A live Bangla voice note becomes a confirmed structured record in under 30 seconds.
SC2	Transcription quality	Word error rate measured and reported on the team’s own labelled test set. Target ≤25%, with the editable transcript covering the remainder.

SC3	Extraction accuracy	≥85% field-level accuracy on cause code against the labelled test set.
SC4	Confirmation speed	Accept-or-correct completes in ≤10 seconds on a mid-range Android device.
SC5	Cluster detection	≥90% of planted correlated-shock scenarios detected, with the false-alert rate stated rather than hidden.
SC6	Offline resilience	Notes recorded with no network sync correctly on reconnect, with no loss and no duplication.

In a field pilot — how it would be proven for real

ID	Criterion	Target and method
SC7	Adoption	≥70% of visits produce a confirmed voice note within 8 weeks of rollout. Adoption is the real risk, so it is measured first.
SC8	Lead time	Median days from first captured signal to a manager action, against the arrears-report baseline.
SC9	Action rate	Percentage of alerts that result in a recorded intervention.
SC10	Portfolio effect	Portfolio-at-risk movement in pilot branches against matched control branches — measured, not claimed.

Why Should You Build This?

A demo that stops the room in ninety seconds. Play a Bangla voice note, watch a structured record appear. Play two more, watch a ward alert fire with taka attached.

Nothing can block you. You record the corpus yourselves on day one — no institutional data, no system access, no partner, no waiting.

It splits cleanly across three people. Audio and ML, backend and clustering, mobile and dashboard — everyone has real work from hour one.

It attaches to BRAC's most profitable business and to a platform that runs in ten countries. This is a module that could genuinely ship, not a demo that dies on Friday.

And it could only be built here. Bangla, field-first, low-literacy-first. No off-the-shelf product does this, and none is coming.